

CHAPTER 9: APPLICATIONS OF INTEGRATION

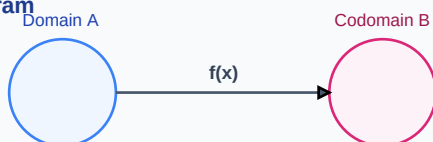
PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 2
Chapter Name:	Applications of Integration

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram



Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Definite Integral Area	The area bounded by a function curve $y=f(x)$, the x-axis, and the ordinates $x=a$ and $x=b$.
Volume of Revolution	The volume of a solid generated by rotating a two-dimensional curve about a coordinate axis.
Gamma Function	An extension of the factorial function to complex and real numbers, defined by an improper integral.

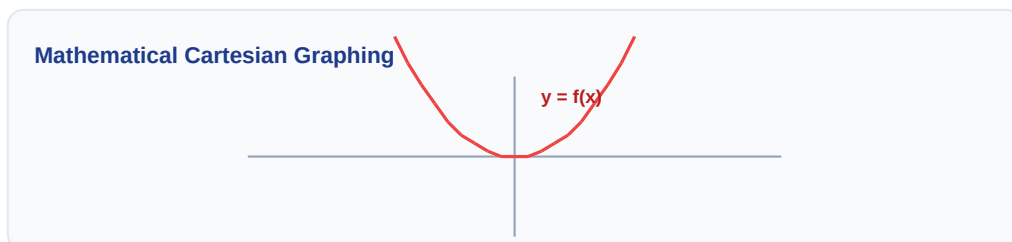
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Area (along x): $A = \int_a^b y \, dx$ | Area (along y): $A = \int_c^d x \, dy$
- Volume (x rotation): $V = \pi \int_a^b y^2 \, dx$
- Gamma Integral: $\Gamma(n) = \int_0^\infty x^{(n-1)} \cdot e^{-x} \, dx = (n-1)!$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find area of the region bounded by $y^2 = 4x$, the x-axis, and lines $x=1$, $x=4$.

Step-by-step Solution:

Area = $\int_1^4 2\sqrt{x} \, dx = 2 \cdot [(x^{1.5})/1.5]_1^4 = (4/3) \cdot (8 - 1) = 28/3$ square units.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the area of the region enclosed between the parabola $y^2 = 4ax$ and the line $y = mx$.

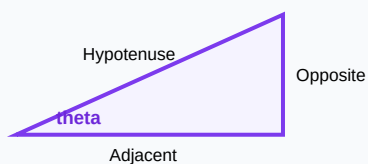
Analytical Solution Strategy:

Intersection: $(mx)^2 = 4ax \Rightarrow m^2x^2 - 4ax = 0 \Rightarrow x = 4a/m^2$. Limits are $x=0$ to $4a/m^2$.

Area = $\int_0^{4a/m^2} [2\sqrt{a}\sqrt{x} - mx] dx = [2\sqrt{a}(x^{1.5})/1.5 - 0.5mx^2]_0^{4a/m^2}$.

Substituting limits yields: $8a^2 / (3m^3)$ square units.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Evaluate definite integral of $\sin^3(x)$ dx from 0 to $\pi/2$ using reduction formula.

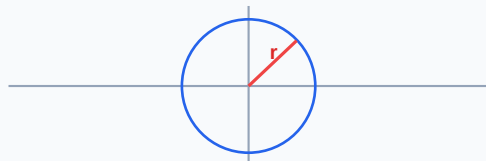
Q2 (2-Mark): Find volume of the solid generated by rotating circle $x^2 + y^2 = r^2$ about x-axis.

Q3 (2-Mark): Calculate area of ellipse $x^2/a^2 + y^2/b^2 = 1$ using integration.

Q4 (5-Mark): Evaluate $\Gamma(5)$.

Q5 (5-Mark): Find area bounded by $y = x^2$ and line $y = 4$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Marine Architecture

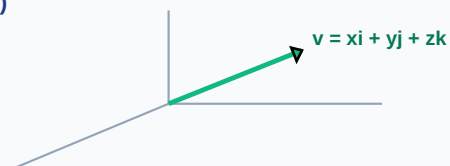
Integrals evaluate the displacement volumes of ship hulls to ensure buoyancy boundaries.

Application Case: Structural Design

Concrete dome construction volumes are calculated using rotation integration rules.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

